



4.2 BASIC CHARACTERISTICS OF TRANSISTORS

MIKROELEKTRONIKA

Selecting the correct transistor for a circuit is based on the following characteristics: maximum voltage rating between the collector and the emitter U_{CEmax} , maximum collector current I_{Cmax} and the maximum power rating P_{Cmax} .

If you need to change a faulty transistor, or you feel comfortable enough to build a new circuit, pay attention to these three values. Your circuit must not exceed the maximum rating values of the transistor. If this is disregarded there are possibilities of permanent circuit damage. Beside the values we mentioned, it is sometimes important to know the current amplification, and maximum frequency of operation.

When there is a DC voltage U_{CE} between the collector (C) and emitter (E) with a collector current, the transistor acts as a small electrical heater whose power is given with this equation:

$$P_C = U_{CE} \cdot I_C.$$

Because of that, the transistor is heating itself and everything in its proximity. When U_{CE} or I_C rise (or both of them), the transistor may overheat and become damaged. Maximum power rating for a transistor, is P_{Cmax} (found in a datasheet). What this means is that the product of U_{CE} and I_C should not be higher than P_{Cmax} :

$$U_{CE} \cdot I_C = P_{Cmax}$$

So, if the voltage across the transistor is increased, the current must be dropped.

For example, maximum ratings for a BC107 transistor are:

$I_{Cmax} = 100mA$,

$U_{CEmax} = 45V$ and

$P_{Cmax} = 300mW$

If we need a $I_C = 60mA$, the maximum voltage is:

$$U_{CE} = \frac{P_{Cmax}}{I_C} = \frac{300mW}{60mA} = 5V.$$

For $U_{CE} = 30V$, the maximum current is:

$$I_C = \frac{P_{Cmax}}{U_{CE}} = \frac{300mW}{30V} = 10mA.$$

Among its other characteristics, this transistor has current amplification coefficient in range between $h_{FE} = 100$ to 450, and it can be used for frequencies under 300MHz. According to the recommended values given by the manufacturer, optimum results (stability, low distortion and noise, high gain, etc.) are with $U_{CE} = 5V$ and $I_C = 2mA$.

There are occasions when the heat generated by a transistor cannot be overcome by adjusting voltages and current. In this case the transistors have a metal plate with hole, which is used to attach it to a heat-sink to allow the heat to be passed to a larger surface.

Current amplification is of importance when used in some circuits, where there is a need for equal amplification of two transistors. For example, 2N3055H transistors have h_{FE} within range between 20 and 70, which means that there is a possibility that one of them has 20 and other 70. This means that in cases when two identical coefficients are needed, they should be measured. Some multimeters have the option for measuring this, but most don't. Because of this we have provided a simple circuit (4.6) for testing transistors. All you need is an option on your multimeter for measuring DC current up to 5mA. Both diodes (1N4001, or similar general purpose silicon diodes) and 1k resistors are used to protect the instrument if the transistor is "damaged". As we said, current gain is equal to $h_{FE} = I_C / I_B$. In the circuit, when the switch S is pressed, current flows through the base and is approximately equal to $I_B = 10\mu A$, so if the collector current is displayed in milliamps. The gain is equal to:

$$h_{FE} = 100 \cdot I_C$$

For example, if the multimeter shows 2.4mA, $h_{FE} = 2.4 \cdot 100 = 240$.

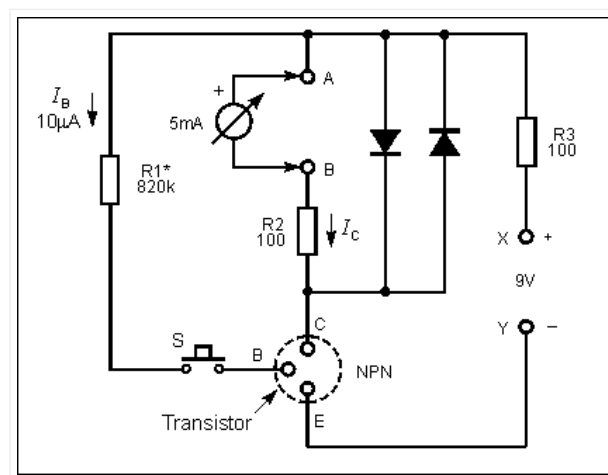


Fig. 4.6: Measuring the h_{FE}

While measuring NPN transistors, the supply should be connected as shown in the diagram. For PNP transistors the battery is reversed. In that case, probes should be reversed as well if you're using analog instrument (one with a needle). If you are using a digital meter (highly recommended) it doesn't matter which probe goes where, but if you do it the same way as you did with NPN there would be a minus in front of the read value, which means that current flows in the opposite direction.



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